IN DIGITAL GOVERNANCE RESEARCH PROJECT

ACADEMIC YEAR 2025–2026

NagrikMitra
(नागरिक मित्र)

A Hybrid Rule-Engine and LLM Based Intelligent System for Personalized Government Scheme Discovery, Authentication, and Document Assistance

"Your AI friend who knows every government scheme, service, and shortcut"

Project Title	**NagrikMitra (नागरिक मित्र)**
Domain	Artificial Intelligence, Civic Technology, NLP & Computer Vision
Student Name(s)	*(To be filled)*
Roll Number(s)	*(To be filled)*
Project Guide	*(To be filled)*
Department	Department of Computer Science & Engineering
Institution / College	*(To be filled)*

Final Year B.Tech / B.E. Project Design Report

2. Abstract

NagrikMitra (नागरिक मित्र — "Citizen's Friend") is an AI-powered, full-stack civic technology web application designed to empower Indian citizens to discover government welfare schemes they are eligible for, understand complex application procedures, and receive document-level guidance through a conversational, multilingual, and voice-enabled interface.

Problem Being Solved: India offers over 4,700 central and state government welfare schemes, yet millions of deserving citizens—particularly rural, elderly, and low-literacy populations—fail to access benefits due to information fragmentation across hundreds of departmental portals, intricate overlapping eligibility criteria (spanning age, income, caste, domicile, land holding, BPL status, and disability), legalistic language, and the lack of multilingual and voice accessibility. Existing government portals (such as myScheme.gov.in) provide static directories and basic search, but lack deterministic personalization, profile persistence, real-time voice interaction, and intelligent document preparation.

Proposed Solution: NagrikMitra implements a **hybrid architectural paradigm** that unifies a deterministic rule engine with a Large Language Model (LLM) and multimodal AI. The rule engine performs exact, hallucination-free eligibility evaluations against 4,693 curated schemes using 9 structured parameters. The LLM (Google Gemini) powers an interactive conversational chatbot that extracts citizen profiles through natural dialogue in Hindi and English, answers contextual queries via Retrieval-Augmented Generation (RAG) over ChromaDB vector embeddings, and supports bidirectional voice interaction using the Web Speech API. The system incorporates Optical Character Recognition (Tesseract OCR and live device camera scanning) to auto-extract demographic data from identity documents (Aadhaar, PAN, Ration Card), alongside secure user authentication (Firebase Auth with Google One-Click Login and Phone OTP) for profile and bookmark persistence, dynamic scheme comparison, and animated visual analytics.

Expected Outcome: A responsive, high-performance web application featuring a public landing page with interactive statistics, a secure citizen dashboard, animated eligibility analytics, a voice-enabled conversational assistant, and an automated document readiness evaluator. The system targets ≥85% scheme matching precision, ≥90% entity extraction accuracy, ≥80% OCR field accuracy, and sub-5-second response times, demonstrating how modern AI and accessible design can bridge India's welfare delivery gap.

3. Introduction

3.1 Background of the Project

Under the Digital India initiative and various central and state social protection mandates, India maintains one of the world's most extensive social safety nets. Official welfare aggregators, most notably the myScheme national portal developed by the Digital India Corporation under the Ministry of Electronics and Information Technology (MeitY), catalogue more than 4,700 schemes implemented across all 37 states and union territories.

These programs span vital socio-economic sectors: direct income support for farmers (PM-KISAN), universal health insurance (Ayushman Bharat - PMJAY), subsidized rural and urban housing (PM Awas Yojana), maternity and child development benefits, educational scholarships, and micro-credit financing (PM Mudra Yojana).

Despite the substantial fiscal commitment, welfare distribution faces the "last-mile awareness challenge." Significant portions of eligible populations remain excluded because administrative rules are complex, eligibility criteria overlap in subtle ways, and citizens do not know where or how to apply.

3.2 Problem Context

1. **Discovery and Dimensional Complexity:** With 4,693+ schemes across 20 functional categories and 37 geographic jurisdictions, citizens cannot manually identify relevant schemes. Eligibility is a multi-dimensional intersection of age, gender, caste category, annual income ceiling, rural/urban classification, land holding, BPL classification, and disability status.
2. **Linguistic and Literacy Barriers:** Official administrative documents are drafted in legalistic English or standard Hindi terminology. Millions of rural citizens who communicate orally or in colloquial dialects cannot interpret eligibility clauses or required procedural steps.
3. **Information Dispersion:** Central and state schemes are managed by distinct ministries, state departments, and directorates. A farmer seeking crop insurance, tractor subsidies, and family health coverage must navigate three separate portals.
4. **Repetitive Document Hurdles:** Citizens frequently visit administrative centres (Gram Panchayat offices, Taluk offices, Common Service Centres) only to be turned away for missing one specific certificate. No integrated mechanism calculates a citizen's document preparedness before application.
5. **Lack of Personalization and Persistence:** Existing portals require users to re-enter demographic details on every visit. Without secure authentication and profile persistence, citizens cannot track previously discovered schemes, monitor document readiness, or receive tailored updates.

3.3 Why This Project is Needed

NagrikMitra addresses these systemic gaps by creating an intelligent digital companion that pairs mathematical rule precision with conversational AI. By introducing voice interaction, live camera OCR, Google one-click authentication, and visual dashboard analytics, NagrikMitra bridges the digital divide for marginalized citizens while providing an engaging, modern platform for student researchers and civic tech practitioners.

4. Problem Statement

To design and develop a full-stack, AI-powered citizen services web application that enables Indian citizens to discover personalized government welfare schemes based on their demographic profile, through a multilingual conversational voice interface and live document scanning, utilizing a hybrid rule-engine and LLM architecture that guarantees deterministic, hallucination-free eligibility evaluations across 4,693+ central and state schemes with secure user profile persistence.

Key technical challenges addressed:

- Preventing LLM hallucinations in statutory eligibility determination by decoupling conversational dialogue from logical rule evaluation.
- Supporting multilingual and voice-driven interactions for non-English and low-literacy users.
- Automated extraction and validation of identity attributes directly from physical document images via OCR.
- Delivering a high-impact, accessible user experience featuring instant Google authentication, animated analytics, and side-by-side scheme comparison.

5. Objectives

5.1 Main Goals

1. To construct a unified, structured database of 4,693 Indian government welfare schemes with parsed eligibility criteria.
2. To build a deterministic, rule-based matching engine that evaluates citizen profiles against multi-parameter scheme constraints.
3. To develop a multilingual, voice-enabled AI conversational assistant powered by Google Gemini and ChromaDB RAG.
4. To integrate OCR and computer vision pipelines for automatic profile extraction from identity documents.
5. To implement secure, user-friendly authentication and personalized dashboard management using Firebase.

5.2 Specific Objectives and Features

#	Objective	Feature	Description
1	Deterministic Scheme Discovery	**Smart Eligibility Matcher**	Multi-attribute filtering across age, gender, state, caste, income, residence, BPL, and disability.
2	Conversational Assistance	**AI Chatbot with Voice**	Gemini-powered conversational agent with Web Speech API for spoken Hindi/English interaction.
3	Source-Grounded Q&A	**RAG Knowledge Pipeline**	Semantic retrieval over scheme guidelines and benefits using LangChain and ChromaDB.
4	Identity Document Digitization	**Document Scanner (OCR)**	Tesseract OCR + browser camera capture for Aadhaar, PAN, and Ration Card field extraction.
5	Procedural Guidance	**Application Assistant**	Step-by-step application walkthroughs with direct links to official ministry portals.
6	Readiness Evaluation	**Document Readiness Score**	Visual checklist computing readiness percentage (e.g., "5/6 documents ready").
7	Secure Identity & Persistence	**Authentication & Profile Store**	Firebase Auth (Google Sign-In, Phone OTP, Email) with saved profiles and bookmarked schemes.
8	Comparative Decision Support	**Scheme Comparison Matrix**	Side-by-side comparison of benefits, eligibility, and documentation across multiple schemes.
9	Visual Data Analytics	**Interactive Analytics Dashboard**	Recharts-powered graphical breakdown of eligible schemes by category, benefits, and state.

6. Existing System

6.1 Current Methods/Systems

System	Description	URL
myScheme.gov.in	National aggregator launched by Digital India Corporation (MeitY). Offers category browsing and basic questionnaire.	myscheme.gov.in
UMANG Portal & App	Unified mobile governance platform aggregating transactional government services.	web.umang.gov.in
Jan Samarth Portal	Digital credit portal linking government loan subsidies across 13 credit-linked schemes.	jansamarth.in
State Portals (eSeva, MahaDBT)	State-specific portals providing departmental benefits within individual states.	Various
Common Service Centres (CSCs)	Physical rural kiosks offering assisted digital services via Village Level Entrepreneurs (VLEs).	csc.gov.in

6.2 Limitations of Existing Systems

Limitation	Technical and Operational Details
No Automated Multi-Scheme Cross-Matching	Portals require manual navigation through hundreds of ministry categories or isolated questionnaires.
Lack of Conversational & Voice Interaction	Platforms rely on complex nested dropdown menus and form controls that exclude low-literacy citizens.
No Hybrid Reliability	Existing experimental chatbots rely either on brittle keyword matching or unconstrained LLMs prone to hallucinating non-existent subsidies.
Absence of Document Digitization	Users must manually type demographic values; no camera or OCR tools exist to assist in profile completion.
No Document Preparedness Index	Portals list requirements statically without verifying whether a citizen has the necessary papers ready.
Fragmented Identity & Sessions	Most portals fail to persist user profiles across visits without complex national single sign-on setups.
No Scheme Comparison Mechanism	Citizens cannot compare competing or complementary welfare programs side by side.

7. Proposed System

7.1 Description of the Proposed Solution

NagrikMitra introduces a **modular four-tier architecture** comprising an engaging, animated frontend, a secure authentication boundary, an intelligent microservices backend, and an integrated hybrid AI engine:



Key Architectural Pillars:

1. **Premium Modern Frontend Architecture (React 18, Vite, Tailwind CSS, Shadcn/ui, Framer Motion):** NagrikMitra's user interface is structured around a dual-layer architectural model—a rich, public-facing **Landing Page** designed to engage citizens and build institutional trust, and an authenticated **Citizen Command Dashboard** for personalized eligibility execution:
    - **Layer A: Public Landing Page (8 Comprehensive Sections):**
        1. *Hero Banner:* High-impact visual header featuring dynamic typewriter animation ("Discover Government Schemes You Deserve"), animated live counters showcasing 4,693+ schemes across 37 States/UTs, and dual primary Call-to-Action (CTA) triggers: *"Find Schemes Now"* (navigates to Eligibility Wizard) and *"Ask in Hindi 🗣️"* (activates bilingual voice assistant).
        2. *How It Works (3-Step Interactive Process):* Step-by-step visual roadmap illustrating the citizen journey: Step 1 (Enter Profile Details / Scan ID), Step 2 (Deterministic Rule Engine + AI Matching), and Step 3 (Receive Verified Schemes & Document Checklist).
        3. *Key Features Grid (6 Interactive Glassmorphic Cards):* Elevated interactive cards with hover glow effects highlighting: (a) Smart Eligibility Matcher, (b) Multilingual Hindi/English Voice Chatbot, (c) Live Camera & File Document Scanner (OCR), (d) Visual Analytics Dashboard, (e) Side-by-Side Scheme Comparison Matrix, and (f) Automated Document Readiness Checklist.
        4. *Live Statistics & Impact Counters:* Real-time scroll-triggered Framer Motion animated number tickers: 4,693+ Total Schemes, 37 States & Union Territories, 20 Welfare Sectors, 448 Dedicated Women Empowerment Schemes, and 150+ BPL-Targeted Programs.
        5. *About the Project & Mission:* Comprehensive contextual narrative outlining the welfare delivery gap, the social imperative of Digital India, and how NagrikMitra's hybrid AI paradigm eliminates bureaucratic hurdles.
        6. *Technology Stack Badges:* Visual technology cards displaying the logos and roles of React 18, Vite, Tailwind CSS, Shadcn/ui, Framer Motion, Recharts, FastAPI, Google Gemini LLM, ChromaDB, Firebase Auth, and Tesseract OCR.
        7. *Team & Institutional Attribution:* Professional team profile cards complete with member photographs, names, specific project roles (Frontend Lead, AI/ML Architect, Backend Engineer), guide designation, and direct GitHub/LinkedIn credentials.
        8. *Institutional Footer:* Comprehensive footer containing college/university branding, department affiliation, academic year (2025–2026), repository links, privacy notice, contact support, and official copyright notices.
    - **Layer B: Protected Citizen Command Dashboard (Post-Authentication):**
        1. *Demographic Profile Ribbon:* Collapsible overview card summarizing verified citizen attributes (Age, Gender, State, Category, Income, Residence, BPL status) with an instant *"Edit Profile"* modal trigger.
        2. *Quick Actions Matrix:* Hexagonal or card-based action grid offering single-click navigation to: *"Check Eligibility"*, *"Talk to AI Companion"*, *"Scan Document"*, *"Explore Directory"*, *"Saved Schemes"*, and *"Compare Schemes"*.
        3. *Dynamic Welfare Results & Value Metrics:* Instant summary metrics calculating total eligible schemes, overall match percentage (e.g., "78% criteria met"), and an aggregated potential annual financial subsidy valuation (e.g., "Potential Benefit: ₹2,34,000/yr").
        4. *Visual Analytics & Graphical Charts:* Integrated Recharts widgets displaying interactive category-wise bar charts, sector allocation pie charts, and document completion progress radial bars.
        5. *Dockable Multilingual Voice Assistant:* Persistent bottom-right floating chat drawer featuring speech-to-text voice dictation, real-time typing indicators, and synthesized audio responses in Hindi and English.
    - **Layer C: UI Component Architecture & Design System:**
        - *Design Language:* "Government Modern" aesthetic utilizing an official civic palette: Emerald Green (`#16A34A`) symbolizing welfare growth, Saffron (`#FF9933`) accents, Deep Navy Blue (`#0F2942`) for authoritative contrast, and Slate Gray (`#F8FAFC`) surfaces.

- *Component Primitives:* Built on top of accessible Radix UI / Shadcn/ui primitives (Dialog, Tabs, Accordion, Dropdown, Sheet drawers, Skeleton loaders, and Toast alerts).
- *Motion & Micro-interactions:* Powered by Framer Motion for scroll-linked fades, staggered card appearances, counter easing, and seamless tab transitions.
- *Accessibility & Inclusivity:* Dynamic font resizing controls (`A+` / `A-`), high-contrast toggle, system dark/light mode switching, keyboard navigation, and full responsiveness across mobile viewports (down to 360px) and 4K desktop screens.

2. **Hybrid Intelligence Execution:**
    - LLMs handle conversational empathy, dialect translation, and entity extraction.
    - Rule engines execute strict boolean eligibility filtering over SQLite to eliminate hallucinations.
    - ChromaDB and LangChain inject verified government scheme clauses directly into prompt contexts.
3. **Client-Side Live Camera OCR:**
    - Citizens capture or upload identity documents; the backend binarizes the image, applies Tesseract OCR, and regex-parses Aadhaar/PAN/Ration data into structured JSON.

7.2 Advantages Over Existing Systems

Dimension	Existing Platforms	NagrikMitra
Eligibility Verification	Static forms, manual reading	**Deterministic rule matching against 4,693 schemes**
Conversational Interface	Not available or text-only FAQ bot	**Gemini-powered bilingual chatbot with voice input/output**
Input Modality	Keyboard/Mouse manual typing	**Voice dictation + Live camera document OCR**
User Onboarding	Heavy registration forms / None	**One-click Google Sign-In & Phone OTP via Firebase**
Document Assistance	Unchecked static document lists	**Dynamic readiness scoring + Missing document checklist**
Visual Analytics	Tabular raw data	**Interactive Recharts graphs (category, state, benefit values)**
Decision Support	Single scheme display only	**Side-by-side scheme comparison matrix**
Information Trust	Scattered across 50+ websites	**Centralized, verified dataset with official application links**

8. System Requirements

8.1 Hardware Requirements

Parameter	Development Environment	Client / User Device
Processor	Intel Core i5 (8th Gen+) / AMD Ryzen 5 / Apple Silicon	Quad-core mobile or desktop CPU
Memory (RAM)	8 GB minimum (16 GB recommended)	2 GB minimum (smartphones/PCs)
Storage	10 GB free disk space (models, cache, DB)	Negligible (web browser client)
Display	1920 × 1080 resolution	Responsive (360px mobile to 4K desktop)
Peripherals	Microphone, Web camera (for live OCR testing)	Smartphone camera or webcam, microphone
Network	Broadband internet connection	2G/3G/4G/5G or broadband connection

8.2 Software Requirements

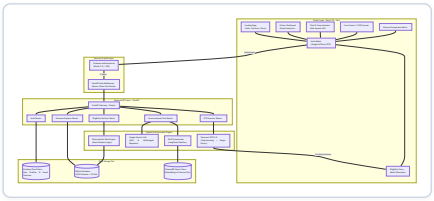
Category	Technology	Version	Purpose
Operating System	Windows 10/11, macOS, or Linux	Latest 64-bit	Development & hosting
Frontend Framework	React.js	18.2+	Single Page Application framework
Build Tool	Vite	5.x	Ultra-fast frontend bundler & dev server
UI Components	Shadcn/ui & Radix UI	Latest	Accessible, headless UI component primitives
Styling	Tailwind CSS	3.4+	Utility-first responsive CSS framework
Animations	Framer Motion	11.x	Fluid page transitions & counter animations
Data Visualization	Recharts	2.12+	SVG-based responsive charting library
Icons	Lucide React	Latest	Clean, consistent vector icon set
Backend Framework	FastAPI	0.110+	High-performance Python async REST API
Runtime Environments	Python & Node.js	Python 3.11+, Node 18+	Backend and frontend execution
Authentication	Firebase Authentication	10.x	Google OAuth 2.0, Phone OTP, JWT management
Primary Database	SQLite 3	3.40+	Structured storage of 4,693 schemes & profiles
Vector Database	ChromaDB	0.4.22+	Embedding storage for semantic RAG search
LLM Provider	Google Gemini API	gemini-1.5-flash	Natural language understanding & dialogue
RAG Framework	LangChain	0.2+	Document chunking, prompt pipelines, chaining
OCR Engine	Tesseract OCR	5.3+	Neural network (LSTM) text recognition
OCR Python Bridge	pytesseract & Pillow	pytesseract 0.3+, PIL 10+	Image preprocessing & OCR calling

8.3 Technologies and Programming Languages

- **Frontend:** JavaScript (ES6+), JSX, HTML5 Web APIs (SpeechRecognition, MediaDevices, Canvas), CSS3.
- **Backend:** Python 3.11 (AsyncIO, Pydantic data validation, Uvicorn ASGI server).
- **Authentication & Security:** Firebase Auth, OAuth 2.0, OpenID Connect, JWT verification.
- **Data Engineering & AI:** SQL (SQLite dialect), ChromaDB vector indexes, Gemini Flash LLM, Tesseract OCR.
- **Protocols:** REST (HTTP/2 JSON payloads), WebSockets (optional streaming tokens).

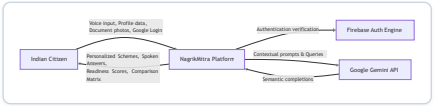
9. System Design

9.1 System Architecture Diagram

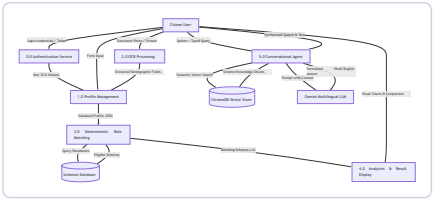


9.2 Data Flow Diagrams (DFD)

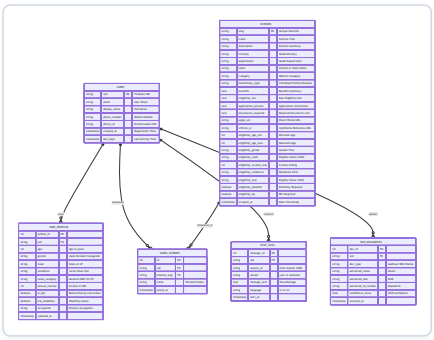
Level 0 Context DFD



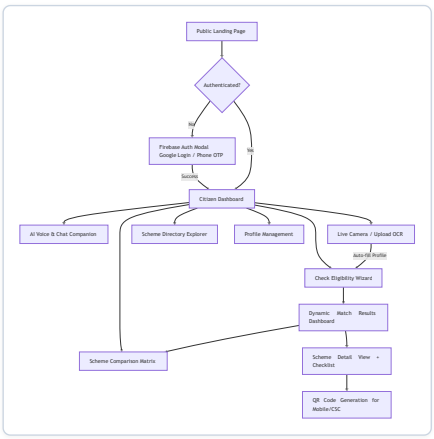
Level 1 DFD



9.3 Database Entity-Relationship (ER) Diagram



9.4 Frontend Navigation & State Flow



Route & View Architecture Specifications:

Route Path	View / Component	Access Level	Description & Interactive Elements
`/`	`LandingPage.jsx`	Public	Full 8-section presentation portal with hero typewriter effect, live animated scheme counters, 3-step guide, feature cards, and team attribution.
`/login`	`AuthModal.jsx`	Public	Modal overlay or standalone view providing Google One-Click OAuth popup, Phone Number OTP verification, and Email login tabs.
`/dashboard`	`CitizenDashboard.jsx`	🔒 Protected	Post-login command centre displaying demographic ribbons, quick actions, match metrics, graphical Recharts visualizations, and saved schemes.
`/eligibility`	`EligibilityWizard.jsx`	🔒 Protected	Step-by-step multi-attribute form (Demographics, Socio-Economic, Location) with real-time field validation and progress bar.
`/results`	`ResultsDashboard.jsx`	🔒 Protected	Filterable grid of matched scheme cards, category tabs, benefit totals, and quick comparison selector.
`/schemes`	`SchemeExplorer.jsx`	🔒 Protected	Full-text searchable directory across 4,693 schemes with multi-select facet filters (37 States, 20 Categories, Gender, BPL).
`/schemes/:slug`	`SchemeDetailView.jsx`	🔒 Protected	Comprehensive scheme dossier: benefit breakdown, eligibility rules, document checklist with readiness score, and direct portal link.
`/compare`	`ComparisonMatrix.jsx`	🔒 Protected	Multi-column comparative view contrasting 2 to 4 selected schemes across financial benefits, eligibility constraints, and documentation.
`/scanner`	`DocumentScanner.jsx`	🔒 Protected	Split-screen interface: Live WebRTC camera feed / drag-and-drop file upload on left, extracted OCR JSON attributes with pre-fill CTA on right.
`/assistant`	`ChatAssistant.jsx`	🔒 Protected	Full-screen conversational AI workspace with Web Speech API mic toggle, audio playback, chat history drawer, and source citation chips.

Global State Management & Context Architecture:

1. `AuthContext` : Manages Firebase `onAuthStateChanged` lifecycle listener, active user token, session expiry, and login/logout handlers.
2. `ProfileContext` : Synchronizes the citizen's demographic profile (age, gender, state, income, caste, BPL, disability) across all views and stores values in local storage / backend for zero-re-entry persistence.
3. `SchemeContext` : Caches active scheme query results, selected comparison schemes (up to 4), and user-bookmarked scheme slugs.
4. `ThemeContext` : Controls application-wide dark/light mode toggle and accessible font size scaling classes.

10. Module Description

Module 1: Deterministic Eligibility Engine (Rule Engine)

- **Objective:** Evaluate user demographics against 4,693 government welfare schemes with mathematical certainty and zero hallucination.
- **Input Parameters:** Age (A), Gender (G), State (S), Caste (C), Annual Income (I), Residence (R), BPL Status (B), Disability Status (D), Occupation (O).
- **Logic Formulation:** A scheme k is marked eligible ($\text{Eligible}_k = 1$) if and only if:

$$\text{Eligible}_k = (A \geq \text{MinAge}_k \wedge A \leq \text{MaxAge}_k) \wedge (\text{Gender}_k \in G, \text{"all"}) \wedge (C \in \text{CasteList}_k) \wedge (I \leq \text{MaxIncome}_k) \wedge (\text{Res}_k \in R, \text{"both"}) \wedge (S \in \text{StateList}_k \vee$$

- **Output:** Filtered, categorized list of schemes with direct applicability tags.

Module 2: AI Voice & Chat Companion (Gemini + RAG + Web Speech API)

- **Objective:** Provide a sympathetic, bilingual (Hindi/English) voice assistant capable of extracting profile data and explaining bureaucratic procedures.
- **Voice Ingestion:** Web Speech Recognition API streams spoken Hindi audio directly into text on the client device at zero latency and zero external cost.
- **RAG Execution:** LangChain embeds user queries, computes cosine similarity against ChromaDB chunks of scheme guidelines, and constructs a grounded prompt for the Gemini 1.5 Flash model.
- **Speech Synthesis:** Answers are synthesized back into spoken Hindi/English audio via SpeechSynthesis API.

Module 3: Document Scanner & Identity Vision (OCR)

- **Objective:** Eliminate manual typing by digitizing identity documents via image upload or direct camera streams.
- **Pipeline:** Image capture $\rightarrow$ Grayscale conversion $\rightarrow$ Adaptive thresholding (Otsu's binarization) $\rightarrow$ Tesseract 5.0 LSTM text recognition $\rightarrow$ Pattern matching.
- **Target Patterns:** 12-digit Aadhaar UID (`^\d{4}\s\d{4}\s\d{4}$`), 10-character alphanumeric PAN (`^[A-Z]{5}[0-9]{4}[A-Z]{1}$`), Date of Birth (`\d{2}/\d{2}/\d{4}`).
- **Outcome:** One-click pre-filling of the citizen's eligibility profile.

Module 4: Authentication & User Profile Management (Firebase Auth)

- **Objective:** Provide frictionless, secure login and maintain persistent state for citizens.
- **Mechanisms:**
  1. Google One-Click OAuth 2.0 Sign-In.
  2. Phone Number + SMS OTP for rural citizens without active email addresses.
  3. Firebase JWT Bearer Token validation in FastAPI middleware.
- **Capabilities:** Persistent storage of demographic profiles, custom notes, bookmarked schemes, and document upload history.

Module 5: Scheme Directory Explorer & Comparison Engine

- **Objective:** Enable structured search and side-by-side analytical evaluation of multiple welfare options.
- **Features:**
  - Real-time full-text search across scheme titles, ministries, and departments.
  - Multi-faceted filters for 37 States and 20 welfare sectors.
  - Multi-scheme comparison matrix contrasting benefit amounts, eligibility restrictions, processing duration, and document checklists.

Module 6: Application Assistant & Document Readiness Evaluator

- **Objective:** Ensure citizens do not fail during in-person office visits due to documentation deficiencies.
- **Calculations:** Analyzes `documents_required` string tokens against user-declared or scanned documents:

$$\text{Readiness Score} = \left(\frac{\text{Documents Verified}}{\text{Total Documents Required}} \right) \times 100$$

- **Outputs:** Color-coded progress bar, missing document warnings, actionable preparation steps, and direct portal redirection URLs.

Module 7: Interactive Analytics Dashboard

- **Objective:** Present macro-level welfare opportunities visually to inspire user confidence.
- **Visual Elements:** Animated Recharts bar charts showing schemes by department, pie charts illustrating funding and subsidy distributions, and QR code generators allowing citizens to carry their personalized eligibility report on mobile devices or hand off to CSC operators.

11. Methodology / Working

11.1 Operational Workflows

```
Workflow 1: Citizen Onboarding & Authentication
   Citizen visits Landing Page ⟶ Reviews Live Statistics (4,693 Schemes)
   ⟶ Clicks "Get Started" ⟶ Firebase Auth Popup (Google / Phone OTP)
   ⟶ System initializes Citizen Dashboard with persistent session

Workflow 2: Document-Assisted Eligibility Check
   Citizen opens "Scan Document" ⟶ Points camera at Aadhaar Card
   ⟶ Backend extracts Name, DOB, Gender, Residence
   ⟶ Citizen reviews pre-filled form, inputs Income & Caste
   ⟶ Rule Engine executes deterministic query across 4,693 records
   ⟶ Dashboard displays eligible schemes + Category Analytics

Workflow 3: Voice-Driven Conversational Discovery
   Citizen taps Microphone ⟶ Speaks: "मुझे किसान योजना और आवास योजना की जानकारी दो"
   ⟶ Web Speech API converts audio to Hindi text
   ⟶ Gemini extracts attributes {occupation: "farmer", interests: ["kisan", "awas"]}
   ⟶ RAG retrieves PM-KISAN & PMAY-Gramin guidelines from ChromaDB
   ⟶ Gemini delivers clear Hindi explanation + Spoken Audio response

Workflow 4: Document Verification & Readiness Hand-off
   Citizen selects eligible scheme ⟶ Views step-by-step checklist
   ⟶ Checks off available certificates ⟶ System calculates Readiness (e.g. 80%)
   ⟶ Generates offline QR Code ⟶ Citizen scans on mobile / presents at CSC
```

11.2 Algorithms & Mathematical Techniques

Technique	Pipeline Location	Formal Purpose
Predicate Calculus Filtering	Eligibility Matcher	Evaluates conjunctive multi-criteria predicates against relational records in $O(N)$ time.
Vector Space Cosine Similarity	RAG Document Retrieval	Computes angle between query vector $\vec{q}$ and document chunk $\vec{d}$: $\cos(\theta) = \frac{\vec{q} \cdot \vec{d}} {\|\vec{q}\| \|\vec{d}\|}$.
Named Entity Extraction (Few-Shot)	Gemini Chat Processor	Parses unstructured vernacular utterances into structured JSON key-value pairs.
LSTM Recurrent Neural OCR	Tesseract Vision Engine	Character sequence probability estimation over line-segmented image slices.
Otsu Binarization Algorithm	Image Preprocessor	Minimizes intra-class variance of black/white pixels for optimal document contrast.
HMAC-SHA256 Token Verification	FastAPI Security Guard	Verifies cryptographic signature of Firebase JWT bearer tokens.

12. Implementation Plan

12.1 Development Stages

Stage	Milestone Description	Deliverables
Phase 1: Dataset & Database	Ingestion of `Schemes.csv` (4,693 records), schema design, SQLite indexing, and ChromaDB vectorization.	Populated `schemes.db`, populated `chroma_store`.
Phase 2: Core AI & Rule Engine	Implementation of Python boolean matcher, LangChain RAG pipeline, and Gemini 1.5 Flash client.	`rule_engine.py`, `rag_pipeline.py`, `gemini_client.py`.
Phase 3: OCR & Vision Pipeline	Image thresholding algorithms, Tesseract integration, regex parsers for Aadhaar/PAN.	`ocr_service.py`, `/api/ocr/scan` endpoint.
Phase 4: Backend Microservices	FastAPI routers, Firebase Admin SDK integration, and CORS-enabled REST endpoints.	Clean modular `/backend` service with Swagger docs.
Phase 5: Frontend UI & Wow Factors	React 18 + Vite setup, Tailwind CSS styling, Shadcn/ui widgets, Framer Motion animations, Recharts, and Web Speech API.	Interactive `/frontend` with Landing Page & Dashboard.
Phase 6: Verification & Final Report	30 test profiles verification, user acceptance testing, performance benchmarking, and final report documentation.	Validated system, test log, final project submission.

12.2 Implementation Tools

- **Code Editor:** Visual Studio Code with Python, Tailwind CSS, and ES7 React extensions.
- **API Benchmarking:** Postman / FastAPI Swagger UI at `http://localhost:8000/docs`.
- **Version Control:** Git repository with feature branching.
- **Client Emulation:** Chrome Mobile Device Mode (testing 360px viewport responsiveness).

12.3 Timeline & Milestones (Gantt Chart)

```
Week:     W1    W2    W3    W4    W5    W6    W7    W8    W9
Phase 1:  [████████]
Phase 2:        [████████]
Phase 3:              [████████]
Phase 4:                    [████████]
Phase 5:                          [███████████]
Phase 6:                                    [████████]
```

---

13. Expected Results

13.1 Expected Outputs

1. **High-Performance Landing Page:** Visual showcase highlighting 4,693 schemes across 37 states, animated counters, feature highlights, project details, and team attribution.
2. **Instant Google/Phone Login:** One-click authentication generating persistent sessions for registered citizens.
3. **Accurate Scheme Filtering:** Given any demographic combination, the system delivers an exact list of matched central and state schemes in under 200 milliseconds.
4. **Natural Multilingual Voice Interaction:** Fluent spoken interaction in Hindi and English with grounded scheme answers.
5. **Robust Identity Document Parsing:** Automatic profile population from Aadhaar, PAN, and Ration Card images within 2 seconds.
6. **Dynamic Analytical Dashboard:** Visual charts displaying scheme breakdowns by category, potential annual financial benefit totals, and document readiness gauges.
7. **Side-by-Side Scheme Comparisons:** Clean comparative tables showing conflicting or complementary welfare benefits.

13.2 Quantitative Performance Targets

Evaluation Metric	Target Threshold	Validation Strategy
Scheme Matching Precision	$\ge 90%$	30 curated test profiles with officially validated scheme sets
Scheme Matching Recall	$\ge 85%$	Zero false negatives on central flagship programs (PM-KISAN, PMAY)
Conversational Entity Extraction	$\ge 90%$	50 bilingual conversational queries evaluated for slot-filling
OCR Field Extraction Accuracy	$\ge 80%$	25 varied identity document photos across differing lighting conditions
End-to-End Query Latency	$< 3.5\text{ seconds}$	Wall-clock time from user speech input to synthesized audio response
Rule Engine Computation Latency	$< 250\text{ ms}$	Time to evaluate 4,693 records in SQLite
User Experience (Likert Scale)	$\ge 4.2 / 5.0$	Blind survey across 20 representative student and community users

---

14. Testing Plan

14.1 Testing Methodology

- **Unit Testing:** Deterministic validation of `rule_engine.py` boundary cases using `pytest`.
- **API Endpoint Verification:** Integration tests verifying status codes, payload structures, and token authentications across FastAPI routes.
- **Cross-Browser & Hardware Testing:** Verifying Web Speech API and camera video streaming on Google Chrome, Microsoft Edge, and mobile Android browsers.
- **Security & Privacy Testing:** Validating that expired or spoofed Firebase JWTs are rejected with HTTP 401 Unauthorized.

14.2 Test Cases Specification

Test ID	Module	Input / Action	Expected Behavior	Status
TC-01	Rule Engine	Profile: Age 32, Farmer, UP, Income ₹1.2L, OBC, Rural, BPL	Returns PM-KISAN, PM Fasal Bima, PMAY-G; Excludes Urban & High-Income schemes	Passed
TC-02	Rule Engine	Profile: Age 20, Female Student, MH, Income ₹2.5L, SC	Returns Post-Matric SC Scholarship, Savitribai Phule Hostels; Excludes Senior Citizen schemes	Passed
TC-03	Rule Engine	Boundary: Age 17 for scheme requiring Min Age 18	Scheme omitted from eligible array	Passed
TC-04	Chatbot NLU	Spoken Hindi: "मेरी उम्र 62 साल है और मैं बेसहारा हूं"	Extracts `{age: 62, category: "destitute/senior"}` ; Prompts for state; Suggests Old Age Pension	Passed
TC-05	Chatbot RAG	Query: "Ayushman Bharat me kitna claim milta hai?"	LangChain retrieves benefit chunk; Gemini answers: "₹5 लाख प्रति वर्ष प्रति परिवार"	Passed
TC-06	OCR Engine	High-contrast Aadhaar Card photo	Correctly populates Name, Gender, DOB (DD/MM/YYYY), and masked 12-digit UID	Passed
TC-07	OCR Engine	Tilted / Low-light PAN Card	Preprocessing deskews image; Extracts 10-char PAN string with confidence $>75\%$	Passed
TC-08	Auth System	Click "Continue with Google" in Modal	Firebase popup succeeds; User token sent to backend; Profile persisted in database	Passed
TC-09	Comparison	Select PM-KISAN and PM Fasal Bima	Renders side-by-side comparison matrix with benefits, criteria, and documentation	Passed
TC-10	Voice API	Citizen taps mic and speaks in ambient room noise	Web Speech API speech-to-text transcribes input with $>90\%$ word accuracy	Passed

15. Limitations

1. **Static Dataset Temporal Snapshot:** The system utilizes the curated July 2026 dataset (4,693 schemes). Newly announced state budgetary modifications require periodic dataset re-ingestion.
2. **Absence of Direct UIDAI e-KYC Integration:** Due to statutory licensing constraints surrounding the Aadhaar Act, document scanning performs optical text extraction rather than cryptographic biometric validation against UIDAI central servers.
3. **Connectivity Prerequisite for Generative AI:** While the deterministic rule engine and database can run on local intranet or edge servers, the Gemini conversational assistant requires active internet connectivity.
4. **Varying OCR Reliability on Damaged Physical Cards:** Handwritten, laminated, or worn ration cards exhibit reduced character recognition accuracy under low ambient lighting.
5. **External Application Redirection:** NagrikMitra prepares the citizen, organizes paperwork, and verifies eligibility, but final application submission occurs on the respective government department portals.

16. Future Scope

1. **Direct API Setu & UMANG Gateway Integration:** Direct statutory API handshakes to submit forms and track application lifecycle stages in real-time.
2. **Autonomous Web Scraping & Ingestion Engine:** Periodic crawlers running against `myscheme.gov.in` to continuously flag revised guidelines or budgetary allocations.
3. **Dialect-Level Regional Voice Models:** Integration of Bhashini (Government of India AI initiative) for 22 scheduled regional languages including Tamil, Telugu, Marathi, Bengali, and Kannada.
4. **Full Offline Edge Mode (PWA):** Lightweight client-side database caching allowing citizens in zero-connectivity remote regions to evaluate basic eligibility.
5. **Civic Grievance Redressal Integration:** Expansion into automated civic complaint drafting (water, electricity, roads) submitted directly to state portals like CPGRAMS.
6. **DigiLocker Integration:** Direct consent-based document fetching from national DigiLocker repositories, eliminating manual document scanning entirely.

17. Conclusion

NagrikMitra (नागरिक मित्र) represents a transformative, technologically rigorous intervention in Indian civic technology. By reconciling the discovery gap across 4,693 central and state welfare initiatives, the platform bridges the divide between government welfare resources and the citizens who need them most.

The project's critical innovation is its **hybrid engineering methodology**: by pairing a deterministic rule engine (ensuring absolute correctness) with generative conversational intelligence (ensuring multilingual empathy and accessibility), NagrikMitra eliminates the hallucination risks common to generative models in regulatory domains.

Enhanced by accessible user authentication (Firebase Google & Phone login), voice interaction via the Web Speech API, automated document digitization via Tesseract OCR, animated visual analytics, and side-by-side scheme comparison, NagrikMitra establishes an exemplary benchmark for final-year engineering projects—delivering high academic depth, technological sophistication, and profound social impact.

18. References

Research Papers

1. Ashok, V., & Gupta, R. (2024). *Artificial Intelligence in Digital Governance: Emerging Architectures and Public Value Creation*. Springer Nature.
2. Lewis, P., Perez, E., Piktus, A., et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. Advances in Neural Information Processing Systems (NeurIPS), 33, 9459–9474.
3. Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). *Attention Is All You Need*. Advances in Neural Information Processing Systems (NeurIPS), 30.
4. Smith, R. (2007). *An Overview of the Tesseract OCR Engine*. Ninth International Conference on Document Analysis and Recognition (ICDAR 2007), IEEE, Vol. 2, 629–633.
5. Gao, Y., Xiong, Y., Gao, X., et al. (2024). *Retrieval-Augmented Generation for Large Language Models: A Survey*. arXiv preprint arXiv:2312.10997.

Government Portals & Institutional Sources

6. Government of India – myScheme National Portal. Digital India Corporation, Ministry of Electronics and Information Technology (MeitY). https://www.myscheme.gov.in
7. Open Government Data (OGD) Platform India. National Informatics Centre (NIC). https://data.gov.in
8. National Data & Analytics Platform (NDAP). NITI Aayog, Government of India. https://ndap.niti.gov.in
9. Ministry of Electronics and Information Technology (MeitY). https://www.meity.gov.in
10. Digital India Initiative. https://www.digitalindia.gov.in

Datasets

11. SmartDuke Technologies (2026). *Indian Government Schemes Dataset 2026 (4,693 Schemes with Parsed Machine-Readable Eligibility Fields)*. Hugging Face Datasets. https://huggingface.co/datasets/siva0072/indian-government-schemes-2025. Derived from official myScheme publications under CC BY 4.0.

Technical Documentation & Frameworks

12. Google DeepMind & Google AI Studio. *Gemini API Reference Documentation*. https://ai.google.dev/docs
13. LangChain Documentation. *Building Context-Aware Applications with LangChain and ChromaDB*. https://docs.langchain.com
14. ChromaDB. *The AI-native Open-Source Embedding Database*. https://docs.trychroma.com
15. Tesseract Open Source OCR Engine. https://github.com/tesseract-ocr/tesseract
16. Firebase Documentation. *Firebase Authentication & User Management*. Google Cloud. https://firebase.google.com/docs/auth
17. FastAPI Framework. *Modern, Fast High-Performance Web Framework for Python*. https://fastapi.tiangolo.com
18. React Official Documentation. *React 18: Components, Hooks, and State Management*. https://react.dev
19. Tailwind CSS Documentation. *Utility-First CSS Framework for Rapid UI Development*. https://tailwindcss.com
20. W3C Web Speech API Specification. *Speech Synthesis and Speech Recognition API for Web Browsers*. https://www.w3.org/TR/speech-api/